Lab 4

BCACC393

Programming with Python Lab

Part A — if-else Based Problems

1. **Electricity Bill**
   An electricity company calculates bills based on the number of units consumed. Write a Python program using if-elif-else to calculate the bill according to different consumption slabs.
2. **Cinema Ticket Pricing**
   A cinema charges different ticket prices based on age. Write a Python program that accepts a person's age and determines the ticket category and price using if-elif-else.
3. **ATM Withdrawal**
   An ATM allows withdrawal only when the account has sufficient balance and the requested amount is a multiple of ₹100. Write a program to check whether a withdrawal can be processed.
4. **Library Fine Calculator**
   A student returns a library book late. The library charges different fines depending on the number of delayed days. Write a Python program to calculate the fine using if-elif-else.
5. **Online Exam Result**
   A university awards grades according to percentage. Write a Python program that accepts a student's percentage and displays the appropriate grade and result status.

Part B — Iteration and Looping: Short Story Problems

6. **The Daily Step Challenge — for Loop**
   Riya is participating in a 7-day fitness challenge. She records the number of steps walked each day. Write a Python program using a for loop to calculate the total number of steps and the average number of steps per day.
7. **The Classroom Attendance — for Loop**
   A teacher wants to record attendance for 10 students. The program should accept P for present and A for absent. Using a for loop, count and display the total number of present and absent students.
8. **The Smart Piggy Bank — for Loop**
   Rahul saves money every day for 10 days. Write a Python program that accepts the amount saved each day using a for loop and calculates his total savings. Also display the highest amount saved in a single day.
9. **The Number Detective — for Loop + if-else**
   A teacher gives a student 20 numbers and asks the student to identify how many are positive, negative, and zero. Write a Python program using a for loop and if-elif-else.
10. **The Cafeteria Counter — for Loop + Dictionary**
    A college cafeteria records the food item ordered by each of 15 students. Write a Python

program using a loop to count how many times each item was ordered and store the result in a dictionary.

11. **The Treasure Hunt — while Loop**
    During a treasure hunt, a player has to enter a secret number. The program should repeatedly ask the player to guess until the correct number is entered. Display the number of attempts made.
12. **The ATM Security System — while Loop**
    An ATM gives a user a maximum of three attempts to enter the correct PIN. Write a Python program using a while loop. The program should allow access if the PIN is correct and lock the account after three incorrect attempts.
13. **The Coffee Machine — Do-While Simulation**
    A coffee machine repeatedly asks the customer to select one of the following options:
    1. Coffee
    2. Tea
    3. Hot Chocolate
    4. Exit

    Python does not have a built-in do-while loop. Write a program that simulates a do-while loop so that the menu is displayed at least once.
14. **The Student Registration Counter — Do-While Simulation**
    A college registration system asks for a student's name and roll number. After registering one student, it asks whether another student should be registered. Write a Python program using a while loop to simulate a do-while structure. At the end, display the total number of registered students.
15. **The Bank Deposit Machine — Nested Loop**
    A bank has 3 customers. Each customer makes 3 deposits during the day. Write a Python program using nested for loops to accept the deposit amounts and calculate the total deposit made by each customer and the overall bank deposit.
16. **The Mini Shopping System — Lists + Dictionary + Loops + if-else**
    A small online store maintains products and prices in a dictionary. A customer can select multiple products and quantities. Write a Python program that repeatedly accepts product names until the customer chooses "checkout". Calculate the total bill and apply a discount using if-else:
    1. Bill >= ₹5000 -> 20% discount
    2. Bill >= ₹3000 -> 10% discount
    3. Bill >= ₹1000 -> 5% discount
    4. Otherwise -> No discount

    Finally, display the purchased items, subtotal, discount, and final bill.
17. **The Morning Exercise Challenge**
    Rohan exercises for 7 days. Each day, he records the number of minutes he exercises. Write a Python program using a for loop to accept the exercise time for each day and calculate the total exercise time.

18. **The Classroom Book Counter**
    A teacher asks 10 students to bring one book each for a classroom library. Write a Python program using a for loop to accept the name of each book and display all the books collected.
19. **The Fruit Basket**
    Mita buys 5 different fruits from a market. Write a Python program using a for loop to accept the name of each fruit and store them in a list. Finally, display all the fruits in the basket.
20. **The Daily Water Tracker**
    A student wants to track how many glasses of water he drinks for 7 days. Write a Python program using a for loop to accept the number of glasses consumed each day and calculate the total number of glasses consumed during the week.
21. **The School Bus Attendance**
    A school bus carries 8 students. The driver wants to record the names of all students travelling on the bus. Write a Python program using a for loop to accept the names of 8 students and display them one by one with their bus seat numbers.

Learning Outcomes

After completing these laboratory exercises, students will be able to:

1. **Apply conditional statements and iteration techniques** such as if-else, for, and while loops to solve computational problems.
2. **Develop problem-solving skills** by designing Python programs based on real-world scenarios and combining multiple programming concepts.